Exam 2 Study Guide

The second exam will have 5 or 6 questions that cover context-free languages. Some questions you will be asked to solve on the exam are:

- Create PDA's for a given language. Study examples shown on Page 115 and 116. Try to generate PDA's for Problems 2.22 through 2.24.
- Create grammars to recognize certain languages. In particular, see example 2.4 on Page 105. Also, see Problem 2.27.
- Create multiple parse trees for ambiguous grammars. Study Problem 2.46 (just show the part that it is ambiguous).
- Show a language is not context-free with the pumping lemma. See Problems 2.31 through 2.33.
- Consider possible variations of the standard single-stack, nondeterminsitic PDA.

Additional practice:

S → AB | C
A → aAb | ab
B → cBd | cd
C → aCd | aDd
D → bDc | bc

Show that this grammar is ambiguous by drawing more than one parse tree. Can you draw three different parse trees? Then, draw an equivalent

PDA.

- Modify the following PDA so that it recognizes the language $\{0^n1^n\} \cup \{0^n10^n\}$.